

Project 3

ELECENG 2EI4 – March 22st, 2026

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Summary

This project involves the design, simulation and implementation of an amplifier that can take an input of $\pm 0.5V$ and $5V$ from a source with an internal resistance of $100\ \Omega$ and deliver it to a $100\ \Omega$ load with good linearity and less than 10% attenuation. This means the output signal of this will yield a $\pm 0.45V$ signal at worst, with the gain of the amplifier as close to 1 as possible. This makes it a buffer amplifier, and will be designed through a common collector circuit using an NPN BJT, which will be simulated and tested using LTSpice and the AD3 respectively.

Design

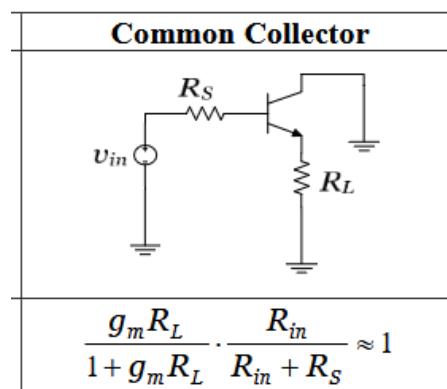
a. What type of transistor did you choose (MOSFET/BJT)? Why?

The transistor used in this circuit is an NPN BJT. This is because the linearity of the BJT is better than the MOSFET. The transconductance (g_m) of a BJT setup will also be higher than the MOSFET when comparing the gain function of both with a common collector and common drain for the BJT and MOSFET respectively.

b. What amplifier topology (CE/CS/CD/etc.) did you choose? Why?

I used a common collector topology, because it is the topology most commonly associated with a voltage buffer, as the goal of this project is to have an amplifier gain of 1. This makes it more suitable compared to a common emitter, which is better for amplifying the voltage, or a common base, which is better suited as a current buffer.

Figure 1: Gain (A_v) equation of a common collector circuit.



c. What calculations did you use to determine the required component values?

Generally for this design with either a BJT or MOSFET, the gain equation is

$A_V = \frac{R_{in}}{R_{in} + R_{sig}} \cdot \frac{R_L}{r_e + R_L}$, where R_{sig} is the source resistance, R_{in} is the load resistance, and r_e is the transistor resistance. Since we use a common collector, we identify that allowed input signal range (or signal swing) is:

$$|V_{in}| = 0.2V_T(1 + g_m R_L)$$

Since we know that V_{in} must be in the 0.5 V range as mentioned in the specification, V_T for a BJT is 25 mV, and R_L is 100 Ω , we can solve for the transconductance g_m :

$$0.5 \leq 0.2(0.025)(1 + 100g_m), g_m \geq 0.99 \text{ mS}$$

For this topology setup, the gain function is $A_V = \frac{g_m R_L}{1 + g_m R_L} \cdot \frac{R_{in}}{R_{in} + R_s}$. Since we know that the $g_m = 0.99$, and both the load and source resistors are 100 Ω , we can rearrange for R_{in} which becomes:

$$0.9 \leq \frac{0.99(100)}{1 + 0.99(100)} \left(\frac{R_{in}}{R_{in} + 100} \right), R_{in} \geq 900 \Omega$$

The equation describing R_{in} in this circuit is a slight modification to the one in the formula sheet for the common collector:

$$R_{in} = R_{eq} \parallel r_{\pi} (1 + g_m R_L)$$

Where R_{eq} is the equivalent resistance of the resistors at the BJT's base, and $r_{\pi} = \frac{\beta}{g_m}$.

Solving for this using the bound for R_{in} previously found:

$$\begin{aligned} 900 \Omega &\leq R_{eq} \parallel r_{\pi} (1 + g_m R_L) \\ 900 \Omega &\leq R_{eq} \parallel r_{\pi} (1 + 0.99(100)) \\ R_{eq} &\geq 988 \Omega \end{aligned}$$

Since R_{eq} is the parallel combination of the external resistors at the BJT's base, the parallel combination of R_1 and R_2 must be greater or equal to 988 Ω . While the resistor value is

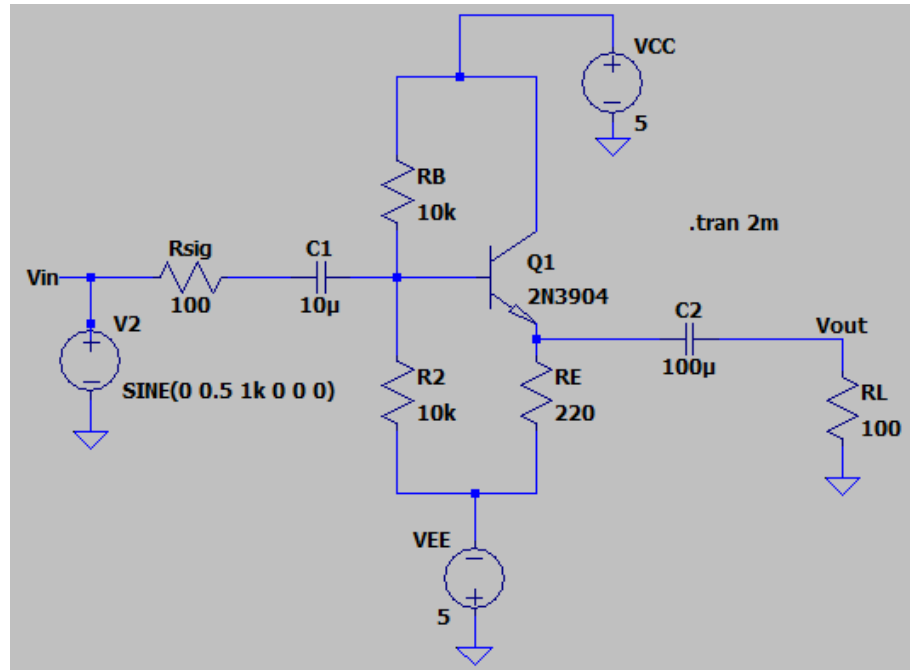
generally negligible over the $988\ \Omega$ constraint, I chose to use two $10\ \text{k}\Omega$ resistors as they were the largest resistors in my kit.

From here, most of the values were negligible as I found through trial and error. The capacitors are generally negligible as the reactance of the capacitors is low in this circuit, but I chose to use the $10\ \mu\text{F}$ and $100\ \mu\text{F}$ capacitors in my kit as their capacitance was high enough to block out the AC frequencies and separate them from the DC biases of the input wave. R_E was found through trial and error as well, as any value from $100\ \Omega$ to $1\ \text{k}\Omega$ kept the output within constraints, so a $220\ \Omega$ resistor was chosen as it gave the best output from the resistors in my kit. I avoided using series or parallel combinations for the R_E as it could impact the output waveform with more tolerances and introduce more noise.

Simulation

Circuit Schematic

Figure 2: Circuit Schematic in LTSpice.



Netlist

```
* C:\Users\adipg\Documents\LTspice\P3AC.asc
* Generated by LTspice 26.0.1 for Windows.
Q1 N001 N003 N004 0 2N3904
VCC N001 0 5
C1 N003 N002 10μ
V2 Vin 0 SINE(0 0.5 1k 0 0 0)
RE N004 N005 220
RB N001 N003 10k
R1 N002 Vin 100
R2 N003 N005 10k
C2 Vout N004 100μ
R3 Vout 0 100
V1 0 N005 5
.model NPN NPN
.model PNP PNP
.lib C:\Users\adipg\AppData\Local\LTspice\lib\cmp\standard.bjt
.tran 5m
.backanno
.end
```

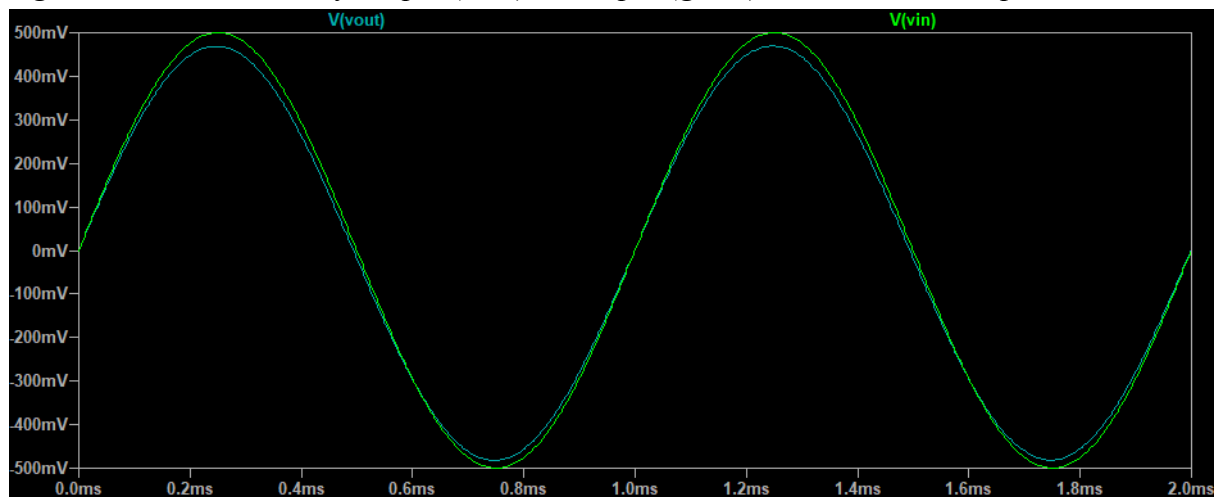
a. How did you model the transistor in the simulator you used?

The LTSpice library has the model for the NPN BJT used in the 2EI4 kit (2N3904), which was imported by changing the model of the NPN component, which copies the characteristics of the BJT into LTSpice.

b. What settings did you use for each simulation (transient/dc sweep/frequency sweep/etc.)?

Transient: A transient sweep is used to compare the output voltage to the input signal as a function of time. This will be configured using the command `.tran 2m`, giving 2 cycles. This will help see the output voltage in comparison to the input, where the voltages at each point can be compared to find the gain A_V .

Figure 3: Transient Plot of Output (blue) and Input (green) as seen in the LTSpice simulation.



$$V_{o, \max} = 0.470 \text{ V}, V_{i, \max} = 0.5 \text{ V}$$

c. The overall gain determined from the simulations.

$$A_V = \frac{V_{out, \max}}{V_{in, \max}} = \frac{0.470 \text{ V}}{0.5 \text{ V}} = 0.94 \geq 0.9$$

$$Attenuation = 100(1 - A_V) = 100(1 - 0.94) = 6\%$$

Therefore, the gain from the simulation is 0.94, and the attenuation is 6%, which is well within the expected range, successfully fulfilling the project's primary design objective for signal preservation while driving a 100Ω load.

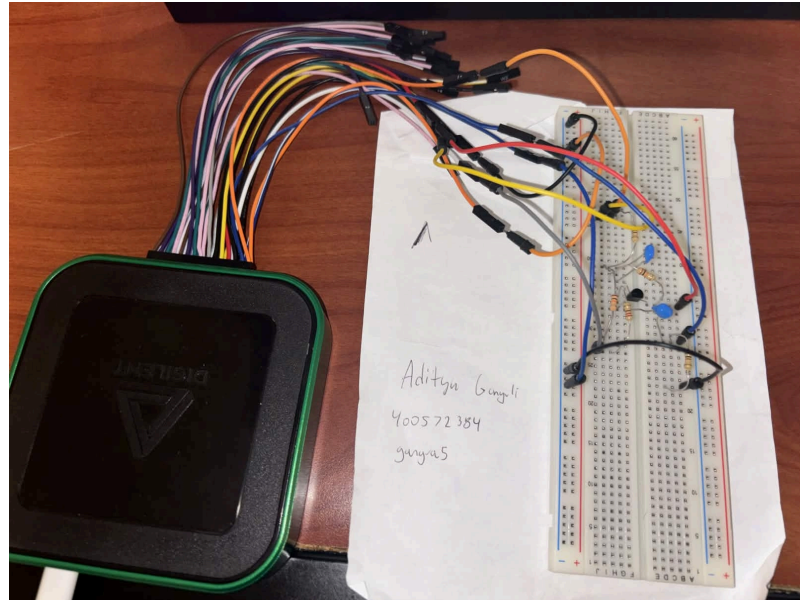
d. Other performance parameters determined from the simulations.

From observing the graph, the phase difference between the waveforms is zero, and the frequency of the waveforms is the same. This tells us that the amplifier is working in a linear fashion, which is outlined in the project specification.

Measurement and Analysis

Circuit Photograph

Figure 4: Breadboard setup using the AD3 with legend below.



Red Wire: $V_{CC} = 5\text{ V}$.

Grey Wire: $V_{EE} = -5\text{ V}$.

Orange Wires: 1+ and 1-, measuring V_{in} .

Blue Wires: 2+ and 2-, measuring V_{out} .

Black Wires: GND.

Yellow Wire: W1, supplying $V_{in} = \text{SINE}(0\ 0.5\ 1\ \text{kHz}\ 0\ 0\ 0)$.

AD3 Measurements

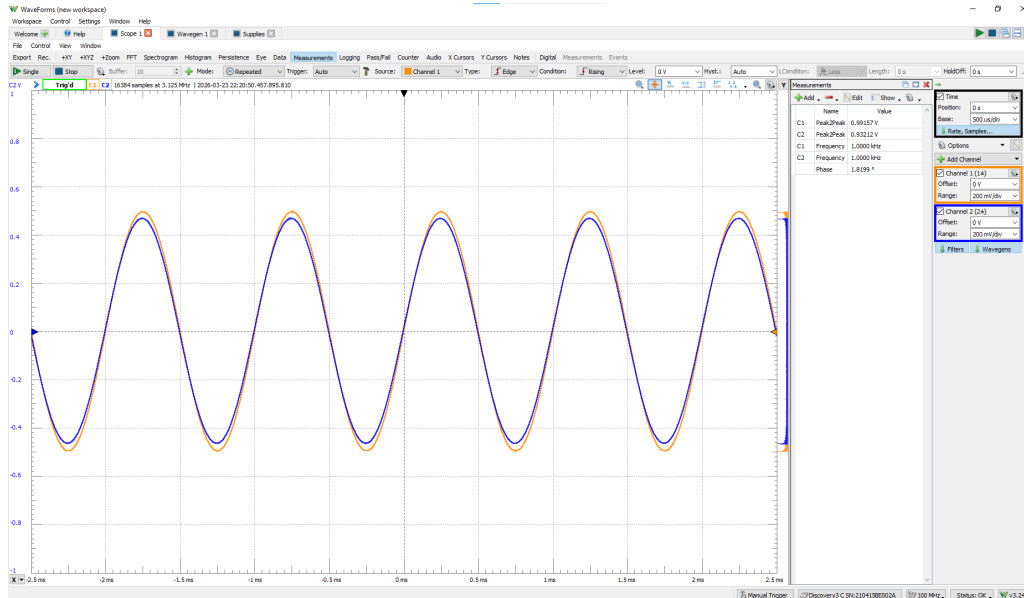
- a. **What measurements did you do? (E.g. waveforms as functions of time, XY plot, network analyzer, etc. You decide which measurements are beneficial and explain why you used them.)**

The measurements were done with the scope feature, which plots the output and input waveforms. Using the measurements tab we can find the peak-to-peak voltage of both channels (output and input), where the ratio between them is the gain. The frequency should be the same and the phase difference should be close to zero.

The Spectrum tool in Waveforms was also used to verify the linearity of the circuit. This is done by checking if any voltage spikes for the input voltage line up with the output, showing that it is linear. These spikes also must be below the 6 dB line, as a spike higher is greater than the 0.5 V outlined in the specification.

- b. How did you determine the midband gain for your amplifier? What value did you obtain? Compare your value with what was expected from calculations and simulations.

Figure 5: AD3 Scope using the Wavegen and the DC supplies for 0.5 V input.



$$V_{o, pp} = 0.93212 \text{ V}, V_{i, pp} = 0.99157 \text{ V}$$

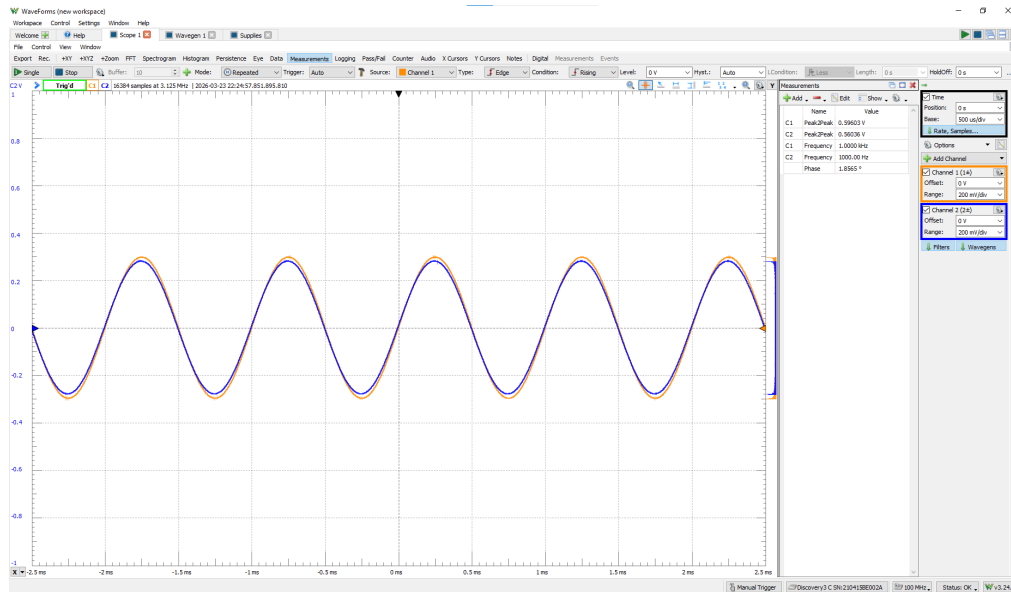
$$A_V = \frac{V_{out, pp}}{V_{in, pp}} = \frac{0.93212 \text{ V}}{0.99157 \text{ V}} = 0.9400 \geq 0.9$$

$$Attenuation = 100(1 - A_V) = 100(1 - 0.9400) = 6\%$$

Based on the measured peak voltages from the simulation, the circuit achieved a midband gain of 0.9400, resulting in a signal attenuation of only 6%. Not only is this in the expected range, but this exactly matches the behaviour of the simulation. Along with this, the frequency is the same at 1 kHz, and the phase difference is minimal, at around 1.82° , which shows that the amplifier functions in a linear manner.

The results are also similar for lower input voltages, such as 0.3 V instead of 0.5 V.

Figure 6: Transient Sweep with 0.3 V input instead of 0.5 V.

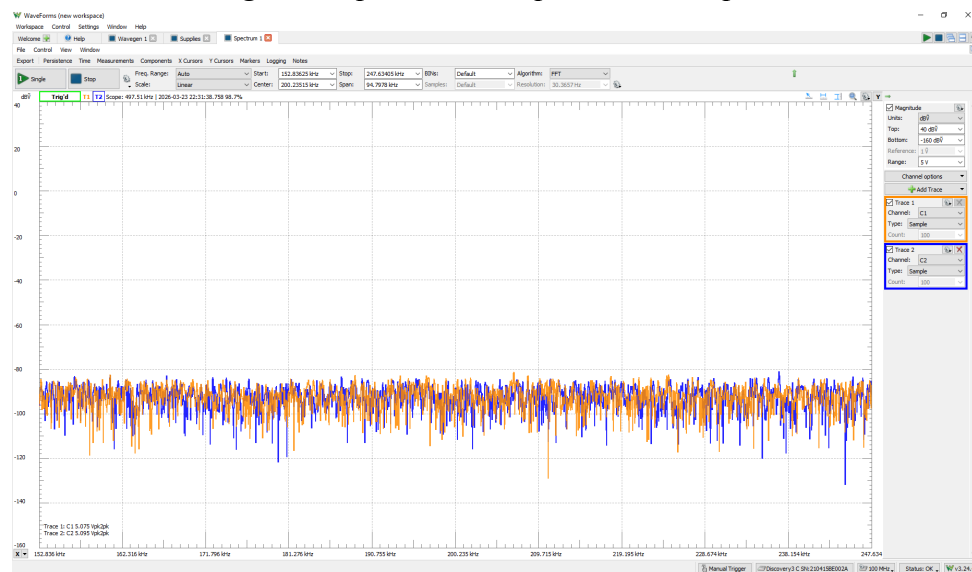


The gain of the amplifier with this input voltage is 0.9401, which is fundamentally the same as with 0.5 V, along with the same frequency and nearly identical phase difference, showing that the common collector circuit is working as an intended buffer.

c. Demonstrate linearity at an input amplitude of 0.5V

As mentioned previously, the waveforms on the Scope with the same frequency and minimal phase difference show that the circuit has great linearity, but this can be further proven using the Spectrum tool in WaveForms.

Figure 7: Spectrum Sweep with 0.5 V input.



Through observing the plot, there are some individual voltage spikes in both the input and output voltages, but these are negligible as the values in dB translate to voltages smaller than 0.5 V. As there aren't any voltage spikes that break past our constraints, we can conclude that this circuit operates in a linear manner with great linearity using both the Scope and Spectrum tools.